

LAB 07

Ex 1:

```
#include<iostream>
using namespace std;
class Calculator{
public:
    int multiply(int a,int b){ return a*b;}
    int multiply(int a,int b,int c){return a*b*c;}
    double multiply(double c,double d){return c*d;}
};
int main(){
    Calculator c;
    cout<<c.multiply(4,3)<<endl;
    cout<<c.multiply(4,3,2)<<endl;
    cout<<c.multiply(4.6,3.3)<<endl;
}
```

Output:

```
15.18
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop,
x1 && "/home/rumaisa-abbasi/Desktop/Lab 07/"ex1
12
24
15.18
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop,
```

Ex 2:

```
#include<iostream>
using namespace std;
class Shape{
public:
    void Draw(){cout<< "Drawing Shape..."<<endl;}
};
class Circle:public Shape{
public:
    void Draw(){cout<< "Drawing Circle..."<<endl;}
};
class Rectangle:public Shape{
public:
    void Draw(){cout<< "Drawing Rectangle..."<<endl;}
};
int main(){
    Shape s; Circle c; Rectangle r;
    s.Draw(); c.Draw(); r.Draw();
}
```

Output:

```
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390
X2 && "/home/rumaisa-abbasi/Desktop/Lab 07
Drawing Shape...
Drawing Circle...
Drawing Rectangle...
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390
```

Task 1:

```
#include<iostream>
using namespace std;
class Calculator{
public:
    void Area(int side){ cout<<"Area of square: "<<4*side<<endl;}
    void Area(int length,int width){cout<<"Area of Rectangle: "<<length*width<<endl;}
    void Area(double radius){cout<<"Area of Circle: "<<3.14*radius*radius<<endl;}
};
int main(){
    Calculator c;
    c.Area(4);
    c.Area(4,3);
    c.Area(2.6);
}
```

Output:

```
cd "/home/rumaisa-abbasi/Desktop/Lab 07/" && g++ Task1.cpp -o Task1 &
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 07$ cd "/home/rumaisa-abbasi/Desktop/Lab 07/"
Task1 && "/home/rumaisa-abbasi/Desktop/Lab 07/"Task1
Area of square: 16
Area of Rectangle: 12
Area of Circle: 21.2264
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 07$
```

Task 2:

```
#include<iostream>
using namespace std;
class Employee{
public:
    void calculateSalary(){cout<< "Calculating Salary..."<<endl;}
};
class Manager:public Employee{
public:
    void calculateSalary(){cout<< "Salary: 20K$"<<endl;}
};
class Developer:public Employee{
public:
    void calculateSalary(){cout<< "Salary: 10K$"<<endl;}
};
class Intern:public Employee{
public:
    void calculateSalary(){cout<< "Salary: 5K$"<<endl;}
};
int main(){
    Employee e; Manager m; Developer d; Intern i;
    e.calculateSalary();
    m.calculateSalary();
    d.calculateSalary();
    i.calculateSalary();
}
```

Output:

```
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/De
Task2 && "/home/rumaisa-abbasi/Desktop/Lab 07/
Calculating Salary...
Salary: 20K$
Salary: 10K$
Salary: 5K$
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/De
```

Task 3:

```
#include <iostream>
using namespace std;
class Complex
{
public:
    int real, imag;
    Complex(int r, int i)
    {
        real = r; imag = i;
    }
    Complex operator + (Complex obj)
    {
        Complex temp(0,0);
        temp.real = real + obj.real;
        temp.imag = imag + obj.imag;
        return temp;
    }
    Complex operator -(Complex obj)
    {
        Complex temp(0,0);
        temp.real = real - obj.real;
        temp.imag = imag - obj.imag;
        return temp;
    }
    Complex operator *(Complex obj){
        Complex temp(0,0);
        temp.real=real*obj.real;
        temp.imag = imag*obj.imag;
        return temp;
    }
};

int main()
{
    Complex c1(2,3); // 2+3i
    Complex c2(4,5); // 4+5i
    Complex c3 = c1 + c2;
    Complex c4=c2-c1;
    Complex c5=c1*c2;
    cout << c3.real << " + " << c3.imag << "i"<<endl;
    cout << c4.real << " + " << c4.imag << "i"<<endl;
    cout << c5.real << " + " << c5.imag << "i"<<endl;
    return 0;
}
```

Output:

```
cd /home/rumaisa-abbasi/Desktop/Lab 07/ &
● rumaisa-abbasi@abduallah-muaz-ThinkPad-L390:~$ g++ 07.cpp -std=c++11 -o 07.exe
● rumaisa-abbasi@abduallah-muaz-ThinkPad-L390:~$ ./07.exe
6 + 8i
2 + 2i
8 + 15i
```

Task 4:

```
#include<iostream>
using namespace std;
class Vehicle{
public:
    virtual void startEngine(){cout<< "Vehicle starting..."<<endl;}
};
class Car:public Vehicle{
public:
    void startEngine(){cout<< "Car starting..."<<endl;}
};
class Bike:public Vehicle{
public:
    void startEngine(){cout<< "Bike starting..."<<endl;}
};
class Truck:public Vehicle{
public:
    void startEngine(){cout<< "Truck starting..."<<endl;}
};
int main(){
    Vehicle v;Car c;Bike b;Truck t;
    v.startEngine();
    c.startEngine();
    b.startEngine();
    t.startEngine();
}
```

Output:

```
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Task4 && "/home/rumaisa-abbasi/Desktop/Lab 07/"Ta
Vehicle starting...
Car starting...
Bike starting...
Truck starting...
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Task4 && "/home/rumaisa-abbasi/Desktop/Lab 07/"Ta
```

